

## PRESENTATION GUIDE

# Smart Air Humidifier

*CpE 421: Emerging Technology · Batangas State University*

This guide walks your group through every slide of the Prototype Presentation, explaining exactly what to put on screen and what the presenter should say. Follow the slide order below for a clear, professional 12–15 minute talk.

| Slide | Title                              | Time     |
|-------|------------------------------------|----------|
| 1     | Title / Cover                      | ~1 min   |
| 2     | Project Overview                   | ~1.5 min |
| 3     | System Components                  | ~2 min   |
| 4     | How It Works (Operating Principle) | ~2 min   |
| 5     | Prototype Demo                     | ~2 min   |
| 6     | Key Features & Functionalities     | ~1.5 min |
| 7     | Testing & Validation Results       | ~2 min   |
| 8     | Impact of Emerging Technologies    | ~1.5 min |
| 9     | Insights & Lessons Learned         | ~1 min   |
| 10    | Conclusion & Q&A                   | ~1 min   |

## SLIDE-BY-SLIDE BREAKDOWN

| SLIDE 1                                                                                                                                                                                                                                                                                                                                                                                                               | TITLE / COVER SLIDE | ~1 min                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose:</b> Introduce the group, project name, and course at a glance.                                                                                                                                                                                                                                                                                                                                            |                     |                                                                                                                                                                                                                                                       |
| <b>What to Include:</b> <ul style="list-style-type: none"><li>• Project title: <b>Smart Air Humidifier</b></li><li>• Subtitle: IoT-Based Automated Air Quality Management System</li><li>• Group number and complete names of all members</li><li>• Course: CpE 421 – Emerging Technology</li><li>• Section, School Year (2nd Sem A.Y. 2025–2026)</li><li>• Batangas State University logo and college name</li></ul> |                     | <b>Visuals / Design:</b> <ul style="list-style-type: none"><li>■ Clean background (dark maroon or white)</li><li>■ BatStateU logo top-center</li><li>■ Project photo or device render as background</li><li>■ Member names in a neat column</li></ul> |
| <b>Tip:</b> The presenter opens with: 'Good morning/afternoon. We are Group 2 and today we will present our prototype — the Smart Air Humidifier.'                                                                                                                                                                                                                                                                    |                     |                                                                                                                                                                                                                                                       |

■ **Tip:** Keep this slide minimal. Let the visuals breathe.

## SLIDE 2

# PROJECT OVERVIEW

~1.5 min

**Purpose:** Give the audience a quick 'What is this?' and 'Why does it matter?'

### What to Include:

- **What it is:** An automated smart humidifier that monitors indoor air quality using an MQ-135 gas sensor
- **Two modes:** (1) Automatic Mode — turns on when odors/VOCs are detected; (2) Manual Mode — controlled via smartphone
- **Problem it solves:** Poor indoor air quality in homes, classrooms, clinics
- **Core technologies used:** ESP32, MQ-135, Relay, Wi-Fi, Web Dashboard
- One-sentence objective from your technical report

### Visuals / Design:

- Short block diagram: Sensor → ESP32 → Relay → Humidifier
- Two icons: Auto Mode vs Phone Control
- Real photo of your finished device

■ **Tip:** Say it in plain language: 'If the air smells bad, the device turns on by itself. You can also control it from your phone.'

■ **Tip:** Avoid jargon here — the overview should be understood by anyone in the room.

## SLIDE 3

# SYSTEM COMPONENTS

~2 min

**Purpose:** Show what hardware and software make the device work.

### What to Include:

- **ESP32 Microcontroller** — brain of the system; Wi-Fi enabled
- **ESP32 Expansion Board** — simplifies wiring and provides stable power rails
- **MQ-135 Gas Sensor** — detects VOCs, smoke, ammonia, CO<sub>2</sub>
- **5V Relay Module (1-channel)** — safely switches humidifier power on/off
- **Printed Circuit Board (PCB)** — permanent, reliable wiring platform
- **Air Humidifier (modified)** — ultrasonic atomizer rewired through relay
- **Software Stack:** C++ / Arduino IDE, MySQL database, Next.js/React dashboard

### Visuals / Design:

- Table or grid with component photo + name + one-line role
- Labeled photo of your actual PCB build
- Software stack icons (Arduino, MySQL, React)

■ **Tip:** Point to each component on the actual device or on a labeled photo as you mention it.

■ **Tip:** For the MQ-135, explain: 'When it smells gas or odor, it sends a signal. The ESP32 reads that signal and decides what to do.'

**Purpose:** Walk through the step-by-step logic of both Auto and Manual modes.

**What to Include:**

- **Phase 1 – Sensing:** MQ-135 continuously reads ambient air quality
- **Phase 2 – Processing:** ESP32 converts analog voltage to an air-quality index
- **Phase 3 – Decision:** Is the reading above the threshold?
- **Auto Mode — YES:** Relay closes → Humidifier turns ON → Mist released
- **Auto Mode — NO:** System stays idle, keeps monitoring
- **Phase 4 – Cooldown:** 5-min mist cycle + 1-min settle delay to avoid false triggers
- **Phase 5 – Reporting:** ESP32 sends live data to web dashboard via Wi-Fi
- **Manual Mode:** User taps button on phone dashboard → HTTP request sent to ESP32 → humidifier toggles regardless of sensor

**Visuals / Design:**

- Flowchart (Figure 1 from your report)
- Two-column layout: Auto Mode vs Manual Mode
- Arrow diagram showing data flow

■ **Tip:** Use your flowchart from the technical report (Figure 1) on this slide — it perfectly illustrates the logic.

■ **Tip:** Emphasize the cooldown: 'We added a settling delay so the mist itself doesn't fool the sensor into thinking air is still dirty.'

**Purpose:** Show the device in action — this is the highlight of the presentation.

**What to Include:**

- Live demo OR pre-recorded video of the device responding to odor (e.g., lighter smoke, alcohol)
- Show the humidifier automatically turning ON when odor is detected
- Show the web/phone dashboard updating in real time
- Demonstrate the Manual Override: toggle on/off from the phone
- Show the cooldown timer activating after the mist cycle
- Display the MySQL database logging sensor readings (screenshot or live)

**Visuals / Design:**

- Live device on the table OR video clip embedded in slide
- Side-by-side: dashboard screenshot + device photo
- Annotated screenshot of web dashboard with labels

■ **Tip:** If doing a live demo, have a lighter or alcohol cotton nearby. Test it before the presentation.

■ **Tip:** If showing a video, keep it under 90 seconds and narrate what is happening on screen.

■ **Tip:** Have a backup screenshot/screen recording in case of Wi-Fi issues.

SLIDE 6

## KEY FEATURES & FUNCTIONALITIES

~1.5 min

**Purpose:** Highlight what makes your device stand out from a plain humidifier.

### What to Include:

- Automated air quality monitoring (MQ-135, real-time)
- Smart automatic activation when odor threshold is crossed
- Wi-Fi connectivity & remote monitoring via web dashboard
- Manual override — phone control from anywhere
- Real-time data logging to MySQL database
- Cooldown & settling mechanism (prevents sensor false triggers)
- Moisture-safe design (electronics physically separated from water tank)
- Affordable open-source solution

### Visuals / Design:

- Icon grid (8 icons, one per feature)
- Screenshot of phone dashboard showing control button
- Comparison table: 'Regular Humidifier vs Smart Air Humidifier'

■ **Tip:** You don't need to read every bullet. Pick the 3–4 features you're most proud of and talk about them naturally.

■ **Tip:** Mention the phone control feature explicitly — it's one of your strongest selling points.

SLIDE 7

## TESTING & VALIDATION RESULTS

~2 min

**Purpose:** Prove that the device actually works correctly and reliably.

### What to Include:

- **Sensor Accuracy Test:** Expose MQ-135 to odor sources (smoke, alcohol) — record if humidifier activated correctly
- **Threshold Test:** Verify the programmed threshold value triggers activation at the right level
- **Auto Mode Test:** Confirm automatic activation and deactivation upon air quality change
- **Manual Override Test:** Confirm phone dashboard successfully toggles the humidifier on/off
- **Cooldown Test:** Verify 5-min mist + 1-min settle cycle works without continuous loop
- **Wi-Fi/Dashboard Test:** Confirm live data updates correctly on the dashboard
- **Data Logging Test:** Confirm sensor readings are stored in the MySQL database
- Note any limitations observed (e.g., MQ-135 sensitivity in humid environments)

### Visuals / Design:

- Test results table (Pass/Fail format)
- Graph of sensor readings vs time (from MySQL logs)
- Photo of odor test being performed

■ **Tip:** Present results as a simple table: Test | Expected Result | Actual Result | Pass/Fail.

■ **Tip:** Be honest about limitations — it shows maturity. Mention the MQ-135 can drift in high humidity.

■ **Tip:** If you ran the device multiple times, show the best average result.

## SLIDE 8

# IMPACT OF EMERGING TECHNOLOGIES

~1.5 min

**Purpose:** Explain which emerging technologies were applied and why they matter.

### What to Include:

- **IoT (Internet of Things):** ESP32 Wi-Fi enables real-time remote monitoring and control — transforms a basic humidifier into a smart device
- **Embedded Systems Automation:** Relay-based automatic actuation eliminates need for manual intervention
- **TinyML (Future):** Could learn room odor patterns and activate mist preemptively — going beyond fixed thresholds
- **Matter Protocol (Future):** Would allow the device to integrate with Google Home or Apple HomeKit ecosystems
- Real-world impact: healthier indoor air in homes, schools, clinics, offices

### Visuals / Design:

- Two columns: Current Tech vs Future Emerging Tech
- IoT ecosystem diagram (Sensor → ESP32 → Cloud → User)
- Logo icons: Matter, TinyML, Google Home, Apple HomeKit

■ **Tip:** The question says 'impact of your chosen emerging technologies' — focus on IoT and automation as your main ones, then mention TinyML and Matter as the next step.

■ **Tip:** Link it to society: 'In the Philippines, indoor air quality is often overlooked. This kind of low-cost smart device can make a real difference.'

## SLIDE 9

## INSIGHTS & LESSONS LEARNED

~1 min

**Purpose:** Share what the team discovered and learned during development.

### What to Include:

- The cooldown/settling mechanism was a critical design decision — without it, the mist itself triggers false positives
- Moisture-safe design (separating electronics from water) is essential for device longevity
- MQ-135 needs warm-up time after power-on before readings stabilize
- Wi-Fi dependency is a limitation — future versions could use local Bluetooth fallback
- Open-source tools (Arduino IDE, Next.js, MySQL) made development affordable and fast
- Physical separation of the control module from the diffuser body was harder than expected but necessary
- Team insights: what each member learned about IoT, embedded systems, and web development

### Visuals / Design:

- 3-column layout: Challenge | What We Did | What We Learned
- Candid photo of the team working on the prototype

■ **Tip:** Be personal here — share a real challenge your group faced and how you solved it. Panels love authenticity.

■ **Tip:** Keep this slide conversational. You don't need to read all the bullets.

## SLIDE 10

## CONCLUSION & Q&A;

~1 min

**Purpose:** Wrap up strongly and open the floor for questions.

### What to Include:

- Restate what the device does in one sentence
- Highlight the two modes: Automatic (MQ-135 triggered) and Manual (phone control)
- Confirm that the prototype was successfully built and tested
- State one future improvement: e.g., TinyML integration or Matter protocol support
- Thank the audience and panel
- Open Q&A: 'We are now open for questions.'

### Visuals / Design:

- Final photo of the complete working prototype
- One-line summary text centered on slide
- 'Thank You' and group member names
- QR code to your web dashboard (optional)

■ **Tip:** End with energy. Say: 'Our Smart Air Humidifier proves that IoT-based solutions don't have to be expensive to be effective.'

■ **Tip:** Assign one member to handle technical questions (hardware), another for software questions.

■ **Tip:** Prepare answers for: 'How accurate is the sensor?', 'What is the range of the phone control?', 'How much did it cost to build?'

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## GENERAL PRESENTATION TIPS

|                           |                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Divide the Slides</b>  | Assign one slide per member if possible. Everyone should speak. Recommended: Slides 1–2 (Member A), Slides 3–4 (Member B), Slide 5 demo (Member C), Slides 6–7 (Member D), Slides 8–10 (Member E or A). |
| <b>Design Consistency</b> | Use Batangas State University colors (maroon and gold) throughout. Keep fonts clean — Helvetica or Arial. Avoid walls of text on any single slide.                                                      |
| <b>The 6×6 Rule</b>       | No more than 6 bullet points per slide, no more than 6 words per bullet. Put the details in what you SAY, not what you TYPE.                                                                            |
| <b>Practice the Demo</b>  | Run the hardware test at least 3 times before the actual presentation. Have a backup video in case Wi-Fi drops or the sensor misbehaves.                                                                |
| <b>Timing</b>             | Target 12–15 minutes total. Leave 3–5 minutes for Q&A.; Practice with a timer.                                                                                                                          |
| <b>Know Your Numbers</b>  | Be ready to cite: sensor threshold value, approximate cost of components, mist cycle duration, and any accuracy percentage from your tests.                                                             |

**Good luck, Group 2! Your Smart Air Humidifier is a well-built, practical IoT project — present it with confidence.**